WORKED SOLUTION

STEP III 2011, Question 5

Pure • Areas by Integration

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/11-s3-q5

This question has a full guided treatment in *Beyond STEP*, Section 4.9, *Holditch's Theorem*: https://beyondstep.keithmcnulty.org/ch04/holditch.html

The key idea

The cartesian formula $(*)$ is useful because it is *bilinear*: if A is P shifted by a vector that depends only on t, the area swept by OA splits into the area swept by OP, an area from the shift alone, and a cross term. The rod gives exactly such shifts, $\mp$ a multiple of the unit vector $(\cos t, \sin t)$. The cross term f looks impossible to evaluate, but we never need to: A and B both go once round the same curve $\mathcal{C}$, so $[A] = [B]$, and that equation pins down f.

Solution

The area formula

With $x = r\cos\theta$ and $y = r\sin\theta$, where r and θ are functions of t, the product rule gives

$$\frac{\mathrm{d}x}{\mathrm{d}t} = \dot{r}\cos\theta - r\sin\theta\,\dot{\theta}, \qquad \frac{\mathrm{d}y}{\mathrm{d}t} = \dot{r}\sin\theta + r\cos\theta\,\dot{\theta},$$

where dots denote $\mathrm{d}/\mathrm{d}t$. So

$$\begin{aligned} x\frac{\mathrm{d}y}{\mathrm{d}t} - y\frac{\mathrm{d}x}{\mathrm{d}t} &= r\cos\theta\,(\dot{r}\sin\theta + r\cos\theta\,\dot{\theta}) - r\sin\theta\,(\dot{r}\cos\theta - r\sin\theta\,\dot{\theta}) \\ &= r^2(\cos^2\theta + \sin^2\theta)\,\dot{\theta} = r^2\frac{\mathrm{d}\theta}{\mathrm{d}t}, \end{aligned}$$

because the $r\dot{r}\sin\theta\cos\theta$ terms cancel. Changing the variable of integration from θ to t,

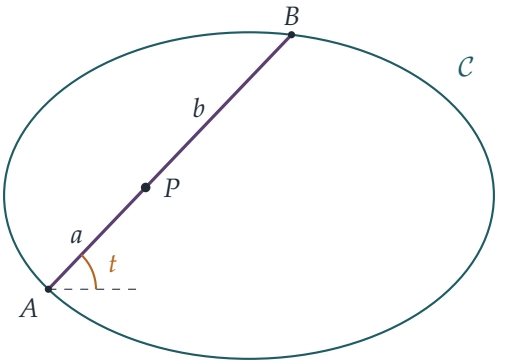
$$\tfrac{1}{2}\int r^2\,\mathrm{d}\theta = \tfrac{1}{2}\int r^2\frac{\mathrm{d}\theta}{\mathrm{d}t}\,\mathrm{d}t = \tfrac{1}{2}\int\left(x\frac{\mathrm{d}y}{\mathrm{d}t} - y\frac{\mathrm{d}x}{\mathrm{d}t}\right)\mathrm{d}t,$$

which is $(*)$.

Coordinates of A and B

The rod points in the direction of the unit vector $\mathbf{u} = (\cos t, \sin t)$, going from A to B, and P lies between them with $AP = a$ and $PB = b$. So A is $P - a\mathbf{u}$ and B is $P + b\mathbf{u}$:

$$A = (x - a\cos t,\ y - a\sin t), \qquad B = (x + b\cos t,\ y + b\sin t).$$



The area swept by OA

Write dots for $\mathrm{d}/\mathrm{d}t$ again. For A we have

$$x_A = x - a\cos t, \quad \dot{x}_A = \dot{x} + a\sin t, \qquad y_A = y - a\sin t, \quad \dot{y}_A = \dot{y} - a\cos t.$$

Expanding,

$$x_A\dot{y}_A = x\dot{y} - a\cos t\,(x + \dot{y}) + a^2\cos^2 t,$$
$$y_A\dot{x}_A = y\dot{x} + a\sin t\,(y - \dot{x}) - a^2\sin^2 t,$$

so

$$x_A\dot{y}_A - y_A\dot{x}_A = (x\dot{y} - y\dot{x}) + a^2 - a\Big[(x + \dot{y})\cos t + (y - \dot{x})\sin t\Big].$$

Now apply $(*)$ with t running from 0 to 2π. The first bracket gives $[P]$, the constant a^2 gives $\frac{1}{2}\cdot 2\pi a^2 = \pi a^2$, and the last term gives $-af$ by the definition of f. Hence

$$[A] = [P] + \pi a^2 - af,$$

as required.

For B, the only change is that $-a$ becomes $+b$ throughout (replace a by $-b$ in the working above). So

$$[B] = [P] + \pi b^2 + bf.$$

The area between $\mathcal{C}$ and $\mathcal{D}$

Ask yourself What are $[A]$, $[B]$ and $[P]$ actually equal to?

As t goes from 0 to 2π the rod turns through one full revolution. Both ends travel anticlockwise once round $\mathcal{C}$ and return to their starting points: after half a turn, A and B have swapped places, and after a full turn they are back where they began. The origin lies inside $\mathcal{D}$, and so inside $\mathcal{C}$. So OA and OB each sweep out the region enclosed by $\mathcal{C}$ exactly once, and OP sweeps out the region enclosed by $\mathcal{D}$ once. Writing $|\mathcal{C}|$ and $|\mathcal{D}|$ for these areas,

$$[A] = [B] = |\mathcal{C}|, \qquad [P] = |\mathcal{D}|.$$

Setting $[A] = [B]$ eliminates $[P]$:

$$\pi a^2 - af = \pi b^2 + bf \quad \Longrightarrow \quad (a+b)f = \pi(a^2 - b^2) = \pi(a-b)(a+b).$$

Since $a + b > 0$ (it is the length of the rod), we may divide to get $f = \pi(a - b)$. Therefore

$$|\mathcal{C}| - |\mathcal{D}| = [A] - [P] = \pi a^2 - a\pi(a-b) = \pi ab.$$

So the area between the curves is πab. Using $[B]$ instead gives $\pi b^2 + b\pi(a-b) = \pi ab$ as well, which is a reassuring consistency check.

Checks

- *A circle.* If $\mathcal{C}$ is a circle of radius R, the chord of length $a + b$ stays at distance $d = \sqrt{R^2 - \frac{1}{4}(a+b)^2}$ from the centre. By Pythagoras, P is at distance $\sqrt{d^2 + \frac{1}{4}(a-b)^2}$ from the centre, so $\mathcal{D}$ is a concentric circle with radius squared $R^2 - ab$. The area between them is $\pi R^2 - \pi(R^2 - ab) = \pi ab$.
- *Symmetry.* The answer is symmetric in a and b, as it must be, because reversing the rod swaps their roles.
- *Numerically.* We moved a rod with $a = 1$, $b = 1.5$ round the ellipse $x^2/16 + y^2/9 = 1$ and computed the area of the traced curve with the shoelace formula. The area between the curves came out as 4.712390, matching $\pi ab = 1.5\pi \approx 4.712389$. We also checked the identities for $[A]$ and $[B]$ by quadrature for randomly chosen curves $P(t)$.

Common slips

- Getting the sign of $\dot{x}_A = \dot{x} + a\sin t$ wrong. Differentiating $-a\cos t$ gives $+a\sin t$.
- Trying to evaluate f directly. It depends on the shape of $\mathcal{D}$ and can't be computed without more information. The point is that $[A] = [B]$ determines it.

Beyond STEP

This is Holditch's theorem (1858): the area between $\mathcal{C}$ and $\mathcal{D}$ depends only on a and b, not on the shape of $\mathcal{C}$. The formula $(*)$ is a special case of Green's theorem, which turns an area integral over a region into a line integral round its boundary. It is also the principle behind the planimeter, a mechanical device that measures area by tracing a boundary. Green's theorem is a centrepiece of first-year vector calculus, and Needham's *Visual Complex Analysis* gives a beautifully geometric view of such boundary integrals.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.